

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	18 June 2026
Team ID	Individual project
Project Name	Registration Ease: Online Complaint Registration and Management System.
Maximum Marks	4 Marks

Technical Architecture:

Technical Architecture – Complaint Ease: Online Complaint Registration and Management System

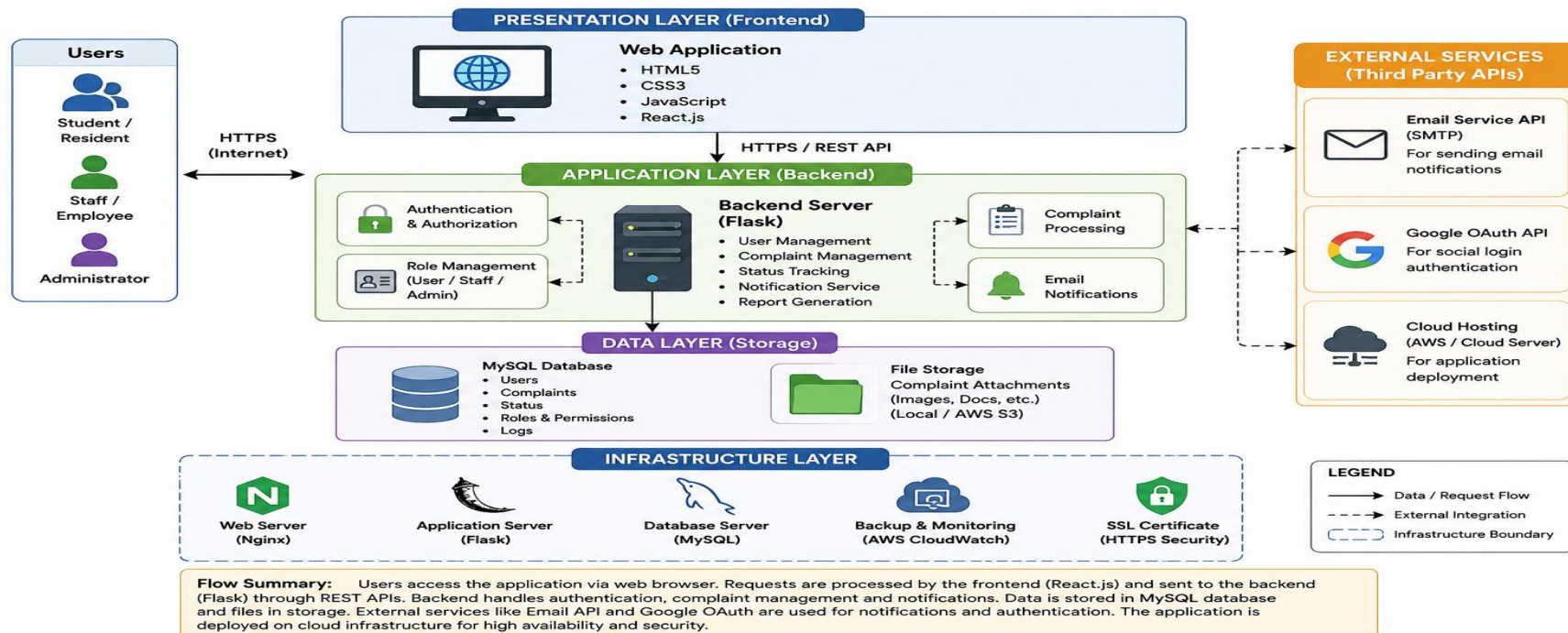


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which users submit and track complaints	HTML, CSS, JavaScript, React.js
2	Application Logic-1	User registration and login processing	Python, Flask
3	Application Logic-2	Complaint registration and complaint submission logic	Flask Backend API
4	Application Logic-3	Complaint management and complaint status update	Python Flask
5	Database	Stores user information, complaint details, status	MySQL
6	Cloud Database	Database backup and cloud storage service	AWS RDS
7	File Storage	Stores complaint attachments/documents/images	Local File System / AWS S3
8	External API-1	Email notification service for complaint updates	SMTP Email API
9	External API-2	User authentication through social login	Google OAuth API
10	Authentication Service	Handles login validation and user authentication	JWT Authentication
11	Web Server	Handles incoming web requests and routing	Nginx
12	Infrastructure	Application hosting and deployment	AWS Cloud / Local Server
13	Monitoring Service	Backup monitoring and system health tracking	AWS CloudWatch
14	Machine Learning Model	Not applicable in current version	N/A

Table-2: Application Characteristics:

	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend and backend development	React.js, Flask
2	Security Implementations	Secure authentication, password encryption, HTTPS protection, access control	JWT Authentication, SHA-256, SSL Certificate
3	Scalable Architecture	System can support increasing number of users and complaints without affecting performance	3-Tier Architecture
4	Availability	Application available continuously with cloud hosting and backup support	AWS Cloud Hosting, CloudWatch Monitoring
5	Performance	Fast complaint submission, quick database response and optimized request handling	MySQL Optimization, API Caching, Nginx